NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Electronics & Communication Engineering

MID SEMESTER EXAMINATION – October 2026

Full Marks:	30	**Course Code:**	EC031304
Course:	Signals and Systems	**Time:**	2 Hours
Semester:	3rd Semester B.Tech.	**Programme:**	Electronics Engineering (VLSI Design and Technology)

(Assume suitable value of missing data, if any.) Answer all questions.

Strictly suggested to attempt the questions in the sequential order only.

Q1. **[1 × 10]** **CO 01** **Level 3**

(a) Find the average power of $\cos(100\pi t) + \sin(200\pi t)$.

(b) Find the period of the discrete-time signal $x(n) = \cos\left(\frac{\pi}{7}n\right) + \sin\left(\frac{\pi}{8}n + \frac{1}{2}\right)$.

(c) Determine the discrete-time Fourier-series coefficients $X(k)$ of $x(n) = \cos\left(\frac{4\pi}{13}n\right) + \frac{1}{2}$.

(d) Determine the Fourier-series coefficients for a periodic square pulse (duty cycle 50%) with two amplitude levels, 1 and 0, and period $T = \pi$.

(e) Consider a system $y(t) = t\,x(t)$. Determine whether the system is bounded.

(f) Find the total energy of the signal

$$x(t) = \begin{cases} 1 + \cos(\omega t), & -\frac{\pi}{\omega} \le t \le \frac{\pi}{\omega}, \\ 0, & \text{otherwise.} \end{cases}$$

(g) Consider the system $y(t) = t^2 x(t-1)$. Determine whether the system is time-invariant.

(h) Consider the system $y(n) = x(n+1) - x(n-1)$. Determine whether the system is linear.

(i) Consider $y(n) = 3x(-n)$. Determine whether the system is causal.

(j) An LTI system is governed by the differential equation

$$\frac{dy(t)}{dt} + 3y(t) = 2x(t).$$

Find the impulse response $h(t)$.

Q2. **[2 × 5]** **CO 02** **Level 1**

(a) Determine the complex-exponential Fourier-series representation of $x(t) = \cos 4t + \sin 6t$. Assume the period of the signal is $T = \pi$.

(b) State and prove Parseval's identity for Fourier series.

(c) State and prove the convolution property of the Laplace transform.

(d) State and prove the frequency-shifting property of Fourier series.

(e) Find the Laplace transform and region of convergence (ROC) of

$$x(t) = e^{-2t}u(t) - e^{-5t}u(t).$$

Q3. **[5 × 2]**

(a) Let $x(t) = e^{-2t}u(t)$ and $h(t) = e^{-3t}u(t)$. Compute the convolution $y(t) = x(t) * h(t)$.

(b) The impulse-response functions of four linear systems S_1, S_2, S_3, S_4 are given below:

$$S_1 : h_1(t) = 1, \qquad S_2 : h_2(t) = u(t),$$
$$S_3 : h_3(t) = \frac{u(t)}{t+1}, \quad S_4 : h_4(t) = e^{-3t}u(t).$$

Here $u(t)$ is the unit-step function. Determine which of these systems are time-invariant, causal, and stable, and justify your answers.

Network Theory – Recreated Circuit Figures

Editable CircuitikZ/TikZ diagrams. Figure geometry has been spaced to reduce label overlap.

Fig. 1

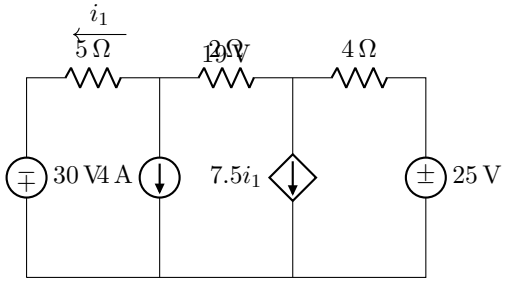


Figure 1: Fig. 1

Fig. 2

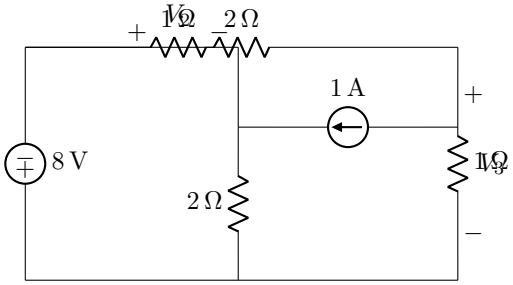


Figure 2: Fig. 2

Fig. 3: Directed graph

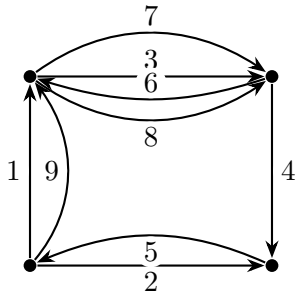


Figure 3: Fig. 3

Fig. 4

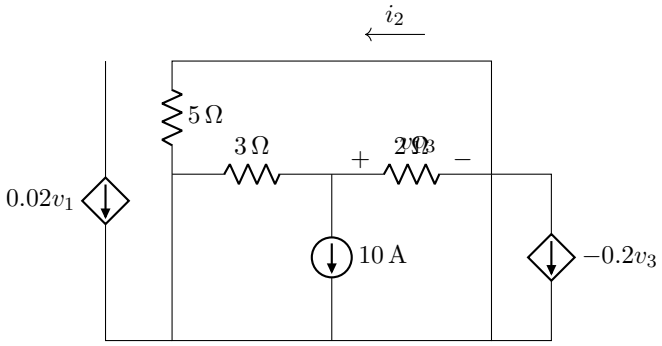


Figure 4: Fig. 4

Fig. 5: Two-port network

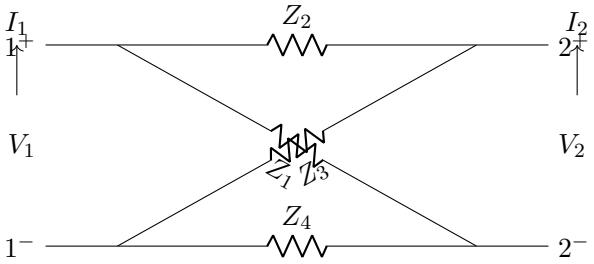


Figure 5: Fig. 5

Fig. 6: Network N

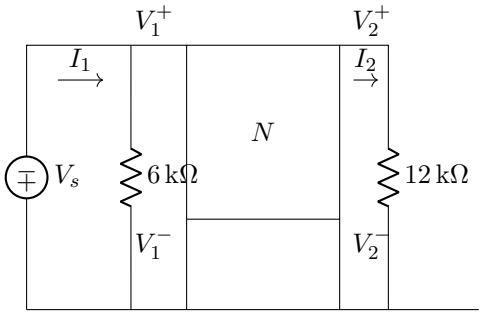


Figure 6: Fig. 6

Fig. 7

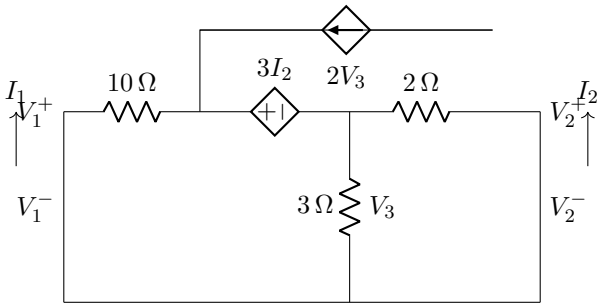


Figure 7: Fig. 7